

# Thermal Engineering

A complete professional handbook for Course 4021. It covers thermodynamics, gas laws, air-standard cycles, IC engine performance, steam, boilers, turbines, heat transfer, heat exchangers and compressors with diagrams, formulas, solved examples, practice questions, model paper and answer guide.

**Code**

4021

**Credits**

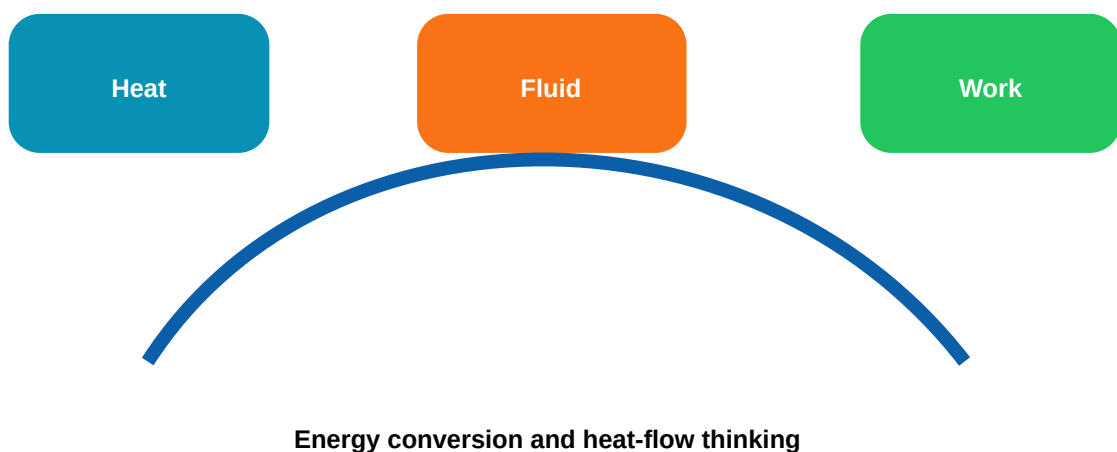
4

**Periods**

60

**Category**

Program Core



# Course Overview

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| Item | Value |
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Diagram + formula + unit + explanation = scoring method.

# Learning Roadmap

CO1 • 16 hours

# 1. Thermodynamics, laws and processes

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

Thermodynamics-□ system boundary, property, heat, work □□□□□ □□□□□□□□□□ □□□□□□□□. Temperature kelvin □□□□□□□ gas equation □□□□□□□□□□□□□□.

scope and application

system, boundary, surrounding

pressure, temperature, enthalpy

Boyle, Charles, Joule and Avogadro laws

gas equation and Mayer relation

heat, work and equilibrium

thermodynamic processes

laws of thermodynamics

entropy and reversibility

P-V and T-S diagrams



Thermodynamics, laws and processes

CO2 • 15 hours

## 2. Air-standard cycles and IC engine testing

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

IC engine testing- $\square$  BP shaft output  $\square\square\square$ , IP cylinder power  $\square\square\square$ , difference friction power  $\square\square\square$ .

air-standard assumptions

Carnot, Otto and Diesel cycles

P-V and T-S diagrams

two-stroke and four-stroke engines

SI and CI engines

valve timing and port timing

IP, BP and FP

thermal efficiencies

SFC, Morse test and heat balance



Air-standard cycles and IC engine testing

CO3 • 15 hours

## 3. Steam, boilers and turbines

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

Steam table question- state first identify . Wet steam  $h = h_f + xh_{fg}$

steam formation

wet, dry and superheated steam

steam tables

Mollier chart

throttling process

boiler classification

Cochran boiler

Babcock and Wilcox boiler

mountings and accessories

impulse and reaction turbines



Steam, boilers and turbines

CO4 • 11 hours

## 4. Heat transfer, heat exchangers and compressors

Textbook chapter with definitions, diagrams, formulas, applications and exam writing method.

Heat transfer-□ mode identify □□□□□□. Conduction solid wall, convection fluid motion, radiation no medium.

conduction, convection and radiation

Fourier law

plane wall conduction

radiation terms

black and grey body

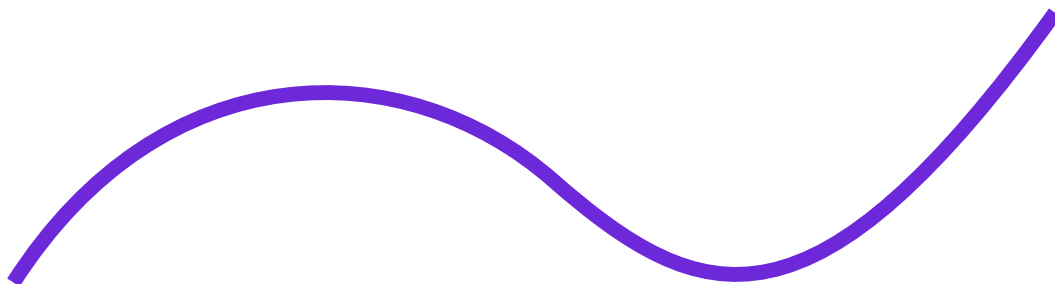
Newton-Richman law

heat exchanger classification

effectiveness

reciprocating compressor

rotary compressor types



Heat transfer, heat exchangers and compressors

## Formula Bank

$pV=nRT$  - ideal gas equation

$C_p - C_v = R$  - Mayer relation

$W = p(V_2 - V_1)$  - constant pressure work

$pV^\gamma = \text{constant}$  - adiabatic process

$\eta_{\text{Otto}} = 1 - T_c/T_h (\gamma - 1)$  - Otto cycle efficiency

$\eta_m = BP/TP$  - mechanical efficiency

$SFC = \text{fuel used per BP hour}$

$h = h_f + xh_{fg}$  - wet steam enthalpy

$Q = kA(T_1 - T_2)/L$  - Fourier law

$\text{effectiveness} = \text{actual heat transfer} / \text{maximum possible}$

## Solved Examples

$$m = pV/RT = 200 \times 0.5 / (0.287 \times 300) = 1.16 \text{ kg.}$$

$$\eta = 1 - 1/8^{0.4} = 56.5\%.$$

$$FP = 5 \text{ kW and } \eta_m = 80\%.$$

$$h = 640 + 0.9 \times 2100 = 2530 \text{ kJ/kg.}$$

$$Q = 0.8 \times 10 \times 30 / 0.2 = 1200 \text{ W.}$$

## Practice and Quick Revision

# Model Paper and Complete Answers

Time: 3 Hours

Max Marks: 100

## Part A

1. Define thermodynamic system.
2. State Zeroth law.
3. Write  $pV=mRT$ .
4. Define air-standard efficiency.
5. Distinguish SI and CI engines.
6. Define BP and IP.
7. Define dryness fraction.
8. Name boiler mountings.
9. State Fourier law.
10. Define heat exchanger effectiveness.

## Part B / C

11. Explain thermodynamic systems.
12. Derive gas equation and Mayer relation.
13. Explain thermodynamic processes with P-V diagrams.
14. Explain first and second laws.
15. Draw and explain Otto and Diesel cycles.
16. Explain petrol and diesel engine working.
17. Explain valve timing and port timing.
18. Explain Morse test and heat balance sheet.
19. Explain steam formation and steam table use.
20. Explain Cochran or Babcock boiler.
21. Compare impulse and reaction turbine.
22. Solve plane wall conduction problem.
23. Explain radiation terms.
24. Compare heat exchangers.
25. Explain single-stage reciprocating compressor.

## References

